BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 03-02-2026**

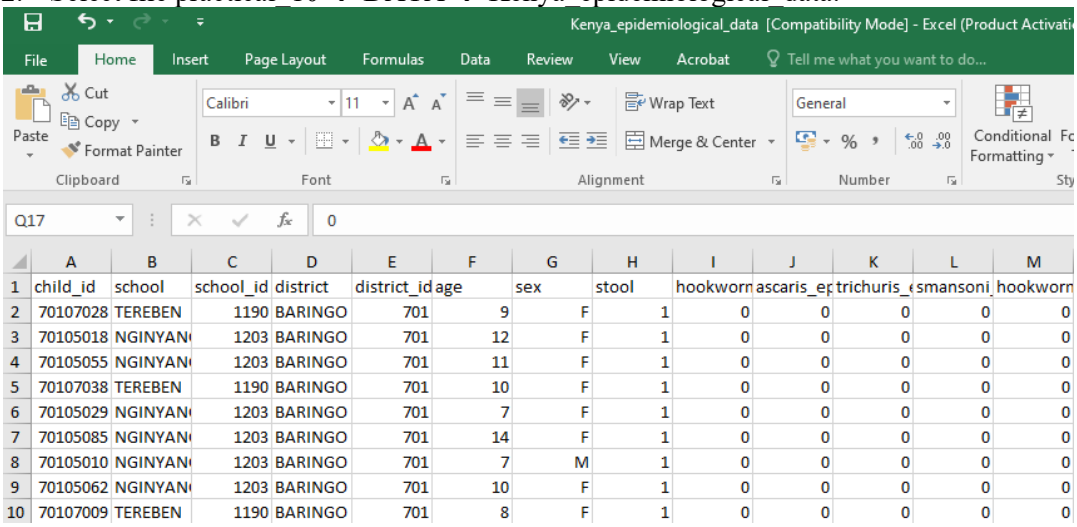
Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 09

Q. Validating map data.

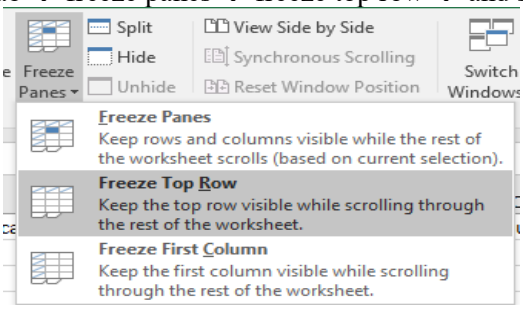
A. Format of the database

1. Download "Practical_10" → open excel.
2. Select file practical_10 → DATA → Kenya_epidemiological_data.

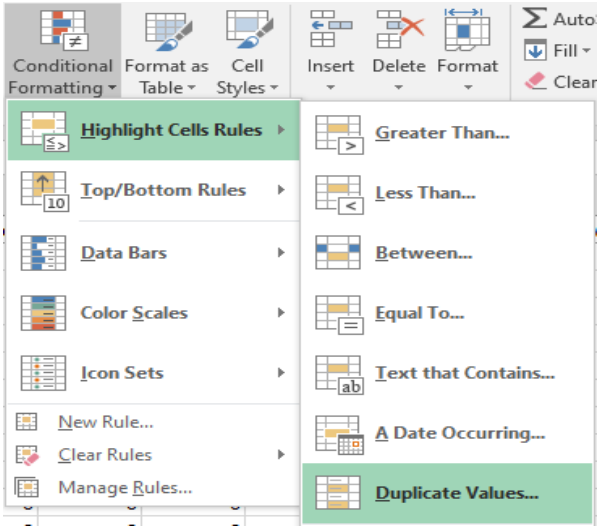


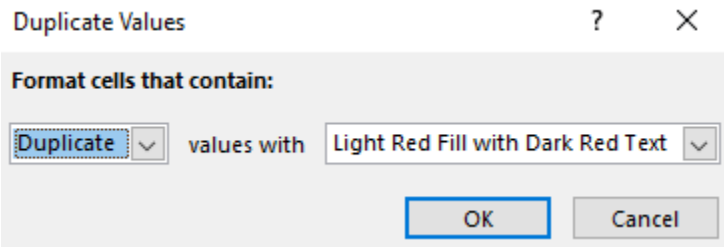
	child_id	school	school_id	district	district_id	age	sex	stool	hookworn	ascaris_ep	trichuris_	smansoni_	hookworn
2	70107028	TEREBEN	1190	BARINGO	701	9	F	1	0	0	0	0	0
3	70105018	NGINYAN	1203	BARINGO	701	12	F	1	0	0	0	0	0
4	70105055	NGINYAN	1203	BARINGO	701	11	F	1	0	0	0	0	0
5	70107038	TEREBEN	1190	BARINGO	701	10	F	1	0	0	0	0	0
6	70105029	NGINYAN	1203	BARINGO	701	7	F	1	0	0	0	0	0
7	70105085	NGINYAN	1203	BARINGO	701	14	F	1	0	0	0	0	0
8	70105010	NGINYAN	1203	BARINGO	701	7	M	1	0	0	0	0	0
9	70105062	NGINYAN	1203	BARINGO	701	10	F	1	0	0	0	0	0
10	70107009	TEREBEN	1190	BARINGO	701	8	F	1	0	0	0	0	0

3. Click on view tab → freeze panes → freeze top row → and select entire child_id (First column)



4. Click on Home tab → conditional formatting → highlight cell rules → duplicate values → select → ok.





Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

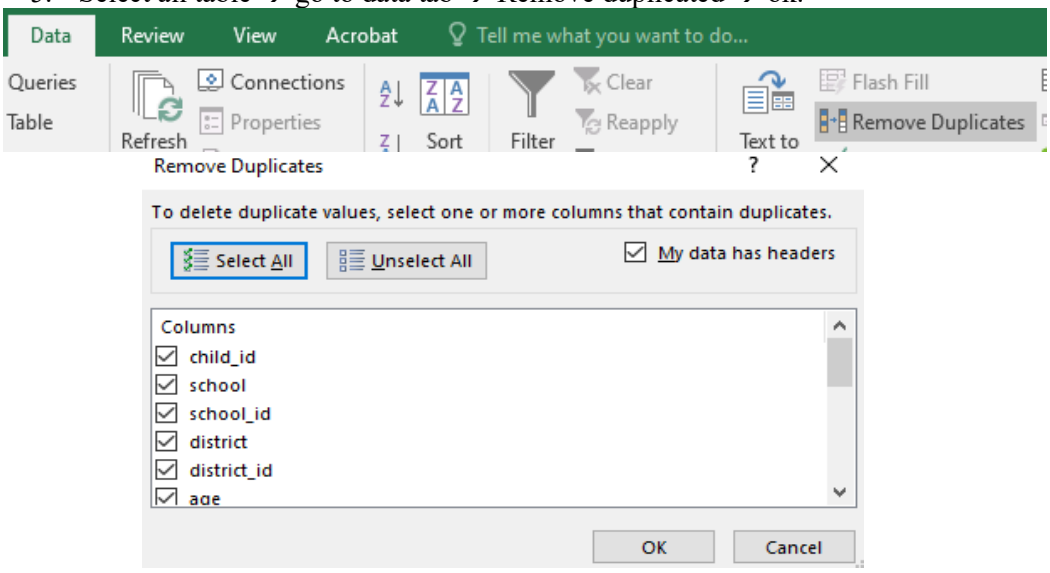
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 03-02-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 09

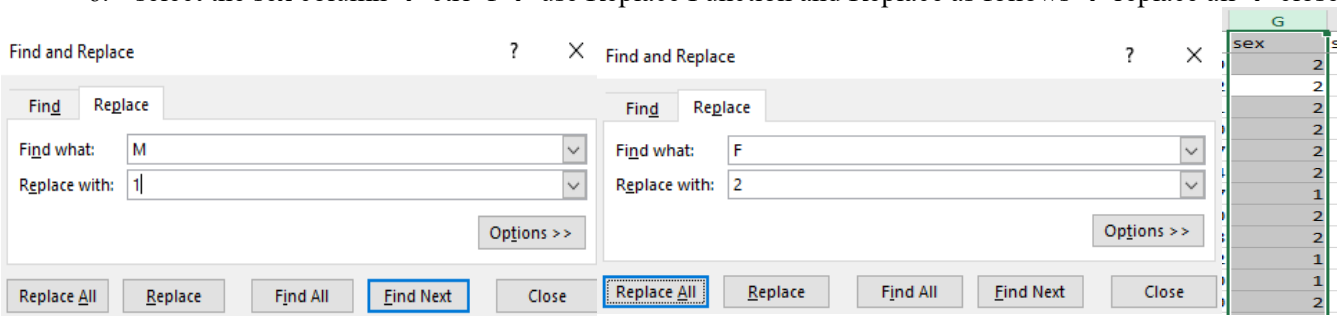
B. Removing Duplicates

5. Select all table → go to data tab → Remove duplicated → ok.

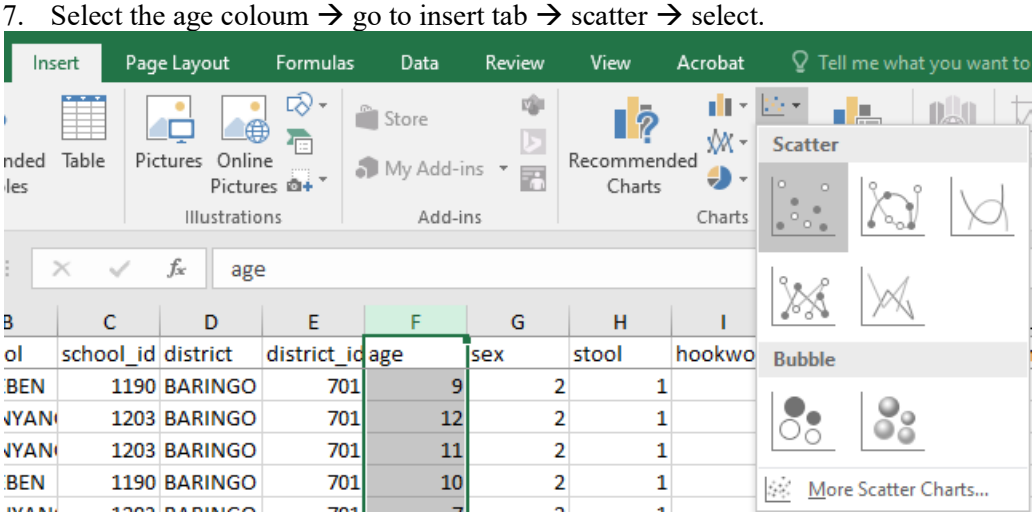


C. Coding of variables

6. select the sex column → ctrl+f → use Replace Function and Replace as follows → replace all → close.



7. Select the age coloum → go to insert tab → scatter → select.



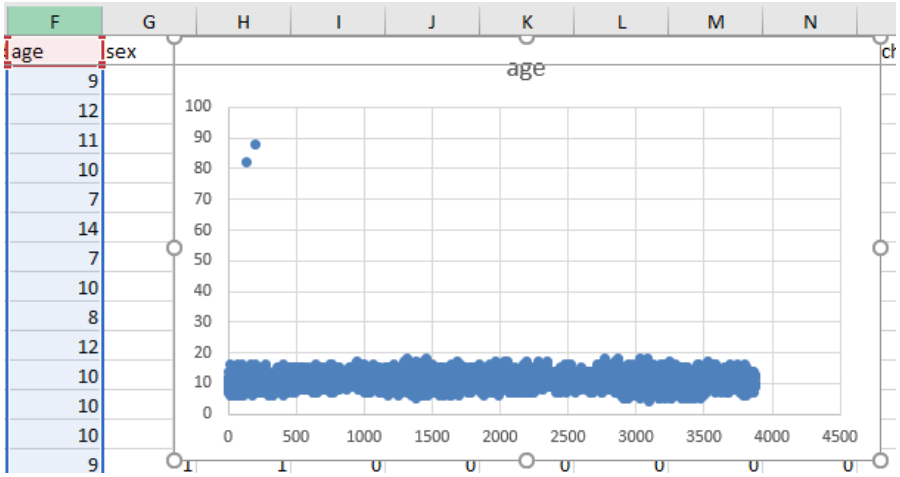
Teacher's Signature

BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 03-02-2026**

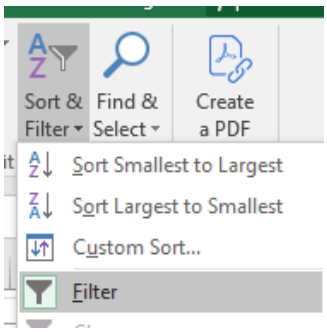
Course Name: Geographic Information System (GIS) **Page no:**

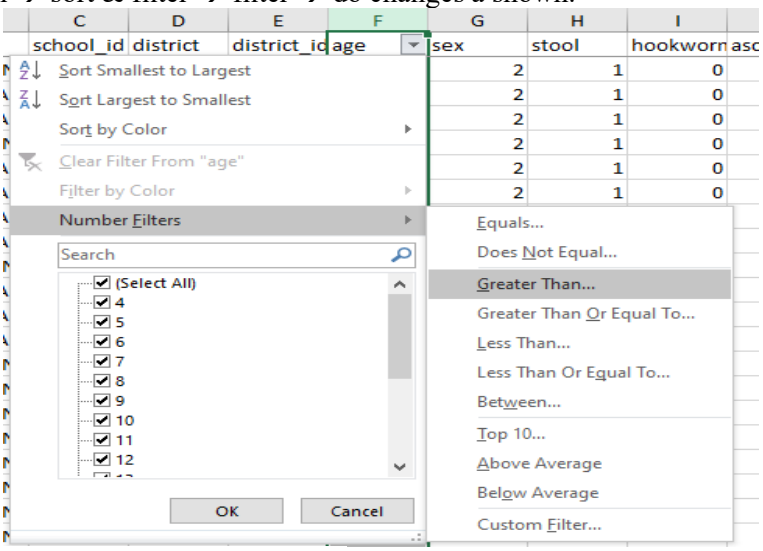
Practical Number:- 09

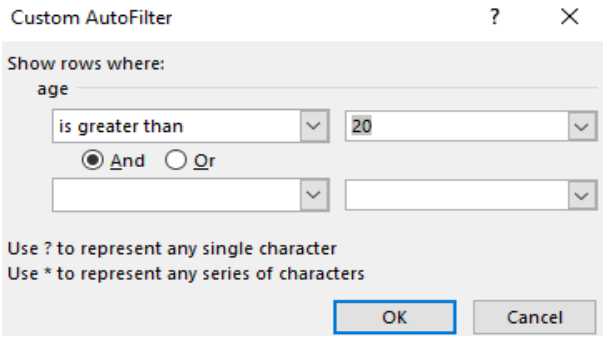


D. Using a filter to detect outliners

8. Home tab → select age coloum → sort & filter → filter → do changes a shown.







Result:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	child_id	school	school_id	district	district_id	age	sex	stool	hookworm	ascaris_ep	trichuris_e	smansoni_	hookworm	ascaris_in	trichuris_i	anysth_in	≤80	
141	70105079	NGINYAN	1203	BARINGO	701	82	2	1	0	0	0	0	0	0	0	0	0	0
202	70203004	SONOKWE	1124	BOMET	702	88	2	1	0	0	0	0	0	0	0	0	0	0

Teacher's Signature

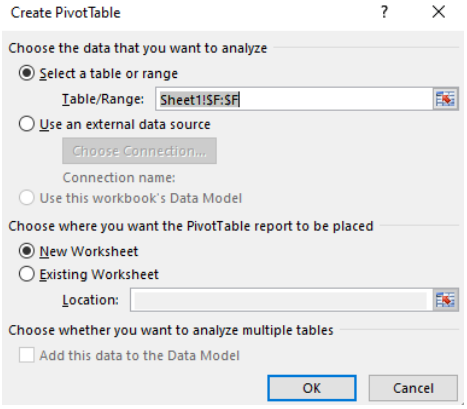
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

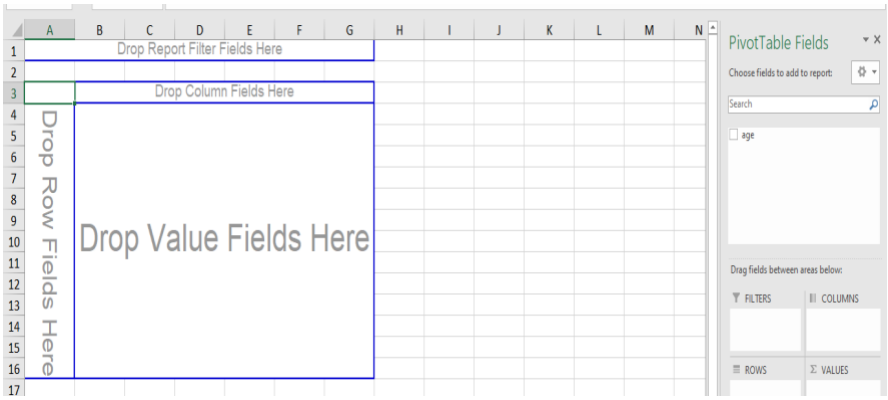
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 03-02-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 09

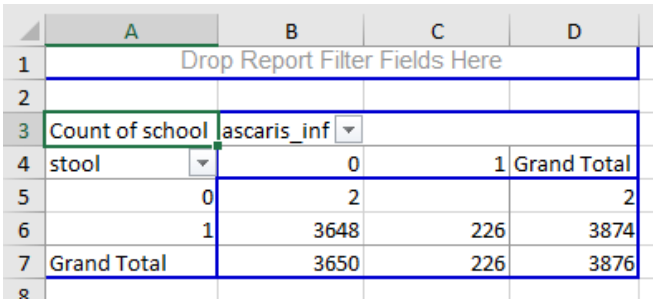
9. Go to insert tap → pivot table → select new worksheet → ok.



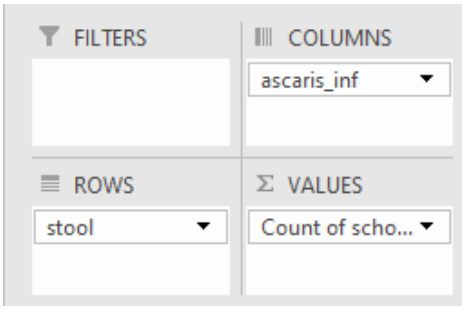


E. Logical Data checks

10. Drag & drop coloum in box → shown in image.

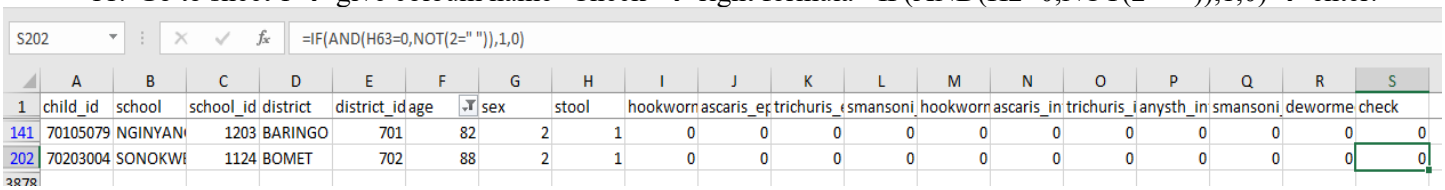


	A	B	C	D
1	Drop Report Filter Fields Here			
2				
3	Count of school	ascaris_inf		
4	stool	0	1	Grand Total
5	0	2		2
6	1	3648	226	3874
7	Grand Total	3650	226	3876



F. Formulas.

11. Go to sheet 1 → give coloum name 'Check' → right formula =IF(AND(H2=0,NOT(2=" ")),1,0) → enter.

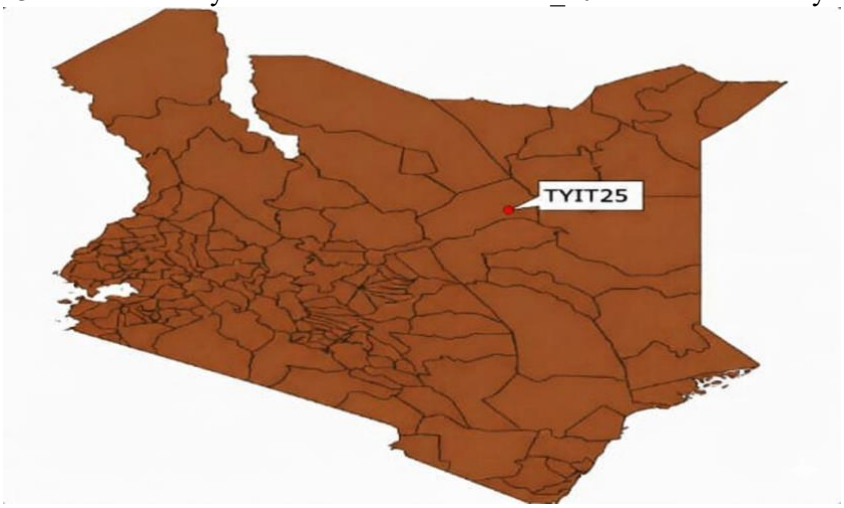


S202 =IF(AND(H63=0,NOT(2=" ")),1,0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	child_id	school	school_id	district	district_id	age	sex	stool	hookworn	ascaris_ep	trichuris_	smansoni	hookworn	ascaris_in	trichuris_i	anysth_in	smansoni	deworme	check
141	70105079	NGINYAN	1203	BARINGO	701	82	2	1	0	0	0	0	0	0	0	0	0	0	0
202	70203004	SONOKWI	1124	BOMET	702	88	2	1	0	0	0	0	0	0	0	0	0	0	0

12. Open QGIS → new project

13. Add vector layer → select file → Practical_10 → DATA → Kenya admin.shp → add → close.



Teacher's Signature

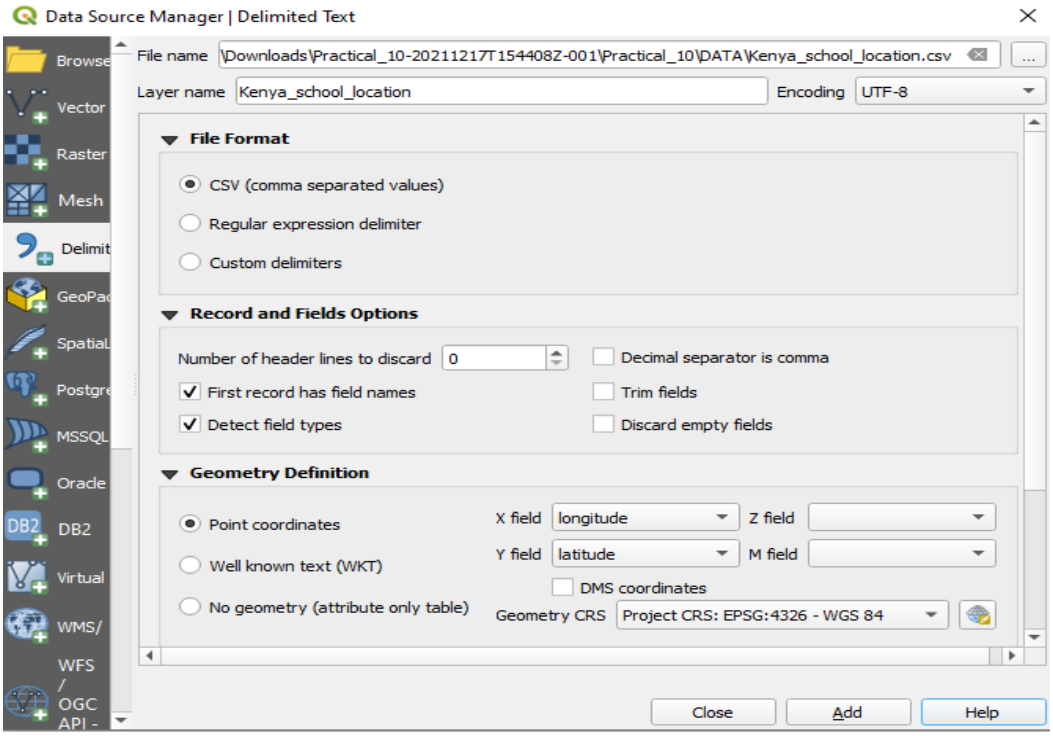
BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

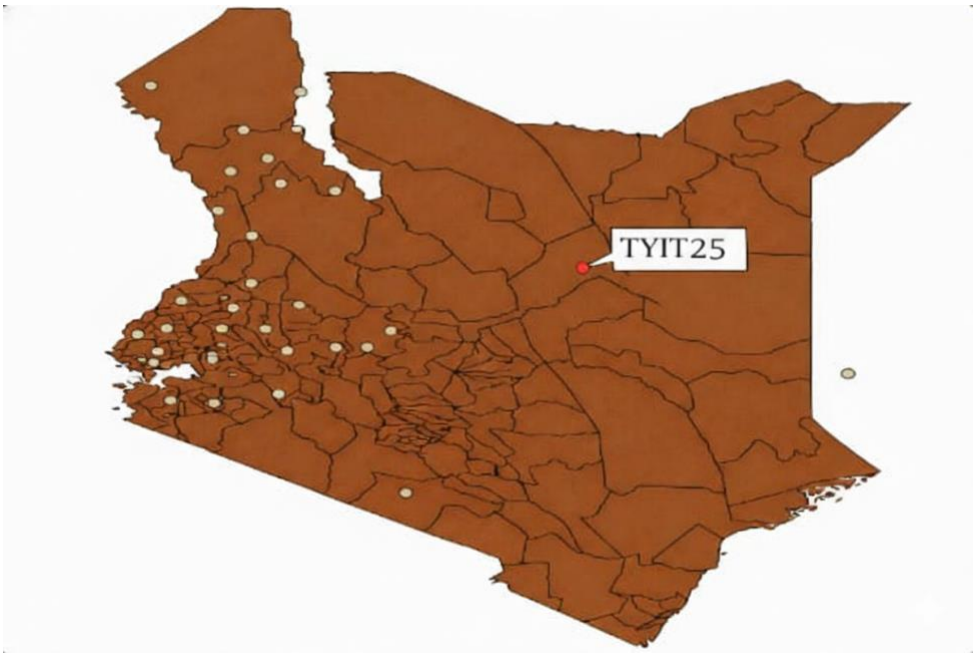
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date: 03-02-2026**

Course Name: Geographic Information System (GIS) **Page no:**

Practical Number:- 09

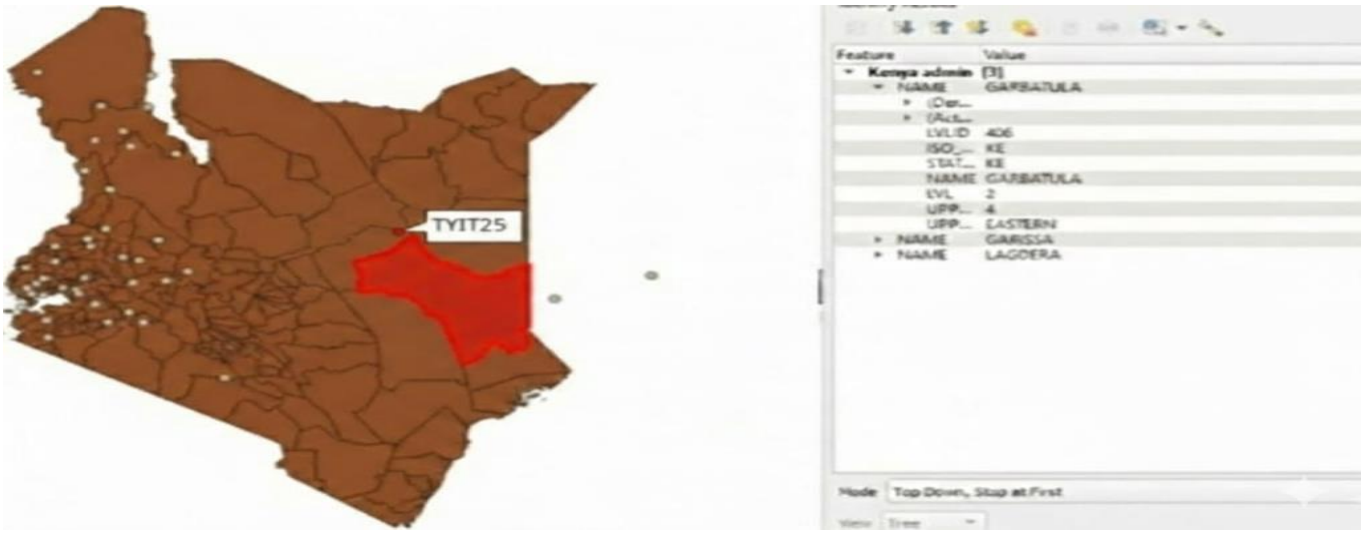
14. Add delimited layer → select file Practical_10 → DATA → Kenya_school_location.csv → do changes → add → close.





Teacher's Signature

15. Then click on identify features.
16. Output:



Teacher's Signature